卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-ECI-DD-EC-0203 30-B003701S-D0203

图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

电气 专业 第 02 卷 第 03 册
Electrical SPECIALITY VOLUME 02 BOOK 03

卷册名称 极2 400V站用电系统及设备安装
Volume name 400V Station Power System and Equipment Installation-Pole 2 System

批 准 APPROVED BY	2019.05.17
审 核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	雷洋 2019.05.17
图 纸 8 张 DRAWING 8 NOS.	打印件 本 PRINTED COPIES NOS.

序号 No.	图 号 DRAWING NO.	文 件 名 称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 Descriptions
1	30-B003701S-D0203-001	卷册设计说明及材料表 Electrical design instructions and material table	A	1		
2	30-B003701S-D0203-002	站用电系统接线示意图 Single Line Diagram of 400V Station Power System	A	1		
3	30-B003701S-D0203-003	极2 400V站用电系统网络图 Network drawing of Pole 2 400V Station Power System	A	1		
4	30-B003701S-D0203-004	极2 400V站用电系统P2A段配置图 Configuration drawing of P2A section of 400V Station Power System	A	1		
5	30-B003701S-D0203-005	极2 400V站用电系统P2B及P2C段配置图 Configuration drawing of P2B and P2C sections of 400V Station Power System	A	1		
6	30-B003701S-D0203-006	极2 400V站用电室布置图 Layout Drawing of Pole 2 400V Station Service House	A	1		
7	30-B003701S-D0203-007	#2继电器室MCC1交流分屏配置及布置图 Configuration and Layout Drawing MCC1 of 500kV No.2 Relay Room	A	1		
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注:

说明：

1. 本卷册设计范围：极2 400V站用电系统、#2 500kV继电器室交流分屏；
2. 本卷册与其余卷册接口：（1）11kV站用电系统详见D0201卷册，11kV干式变进线电缆开列在D0201卷册。（2）公用400V站用电系统详见D0204卷册；（3）极1 400V站用电系统详见D0202卷册；（4）本卷册的动力电缆清册详见D0414卷册；（5）本卷册负责站用变屏至第1个户外检修箱之间的电缆设计，检修箱之间电缆详见D0405卷册；（6）本卷册负责站用变屏至户外照明箱之间的电缆设计，照明箱至各个灯具的电缆详见户外照明相关设计卷册；（7）本卷册负责站用变屏至二次交流电源箱之间的电缆开列，交流电源箱及其后回路设计详见二次专业相关卷册；（8）本卷册负责站用变屏至水工、消防专业相关设备动力柜（箱）的电缆设计，动力柜（箱）及其后回路设计详见相关厂家图纸；（9）本卷册负责站用电屏（或分屏）至建筑物配电箱之间的电缆设计，建筑物配电箱以下的电缆见各建筑电气卷册。
3. 400V站用电系统采用TN-C-S系统，N线在400V交流屏中直接接地。
4. 本卷册中所开列的动力电缆均为阻燃型铠装电缆，电缆敷设需满足D0406~D0410卷册的要求。
5.根据招标结果及厂家提供样本，本站交流400V采用ABB断路器，断路器型号说明如右。

E2.2 S 1250 R1250

- 框架断路器 type of switch
- 分断能力 breaking capability
  - B: 42 kA H: 100 kA
  - C: 50 kA V: 150 kA
  - N: 66 kA X: 200 kA
  - S: 85 kA
- 断路器框架电流 rated current of switch
- 脱扣器额定电流 rated current of tripping device

XT2 S 160 R160

- 塑壳断路器 type of switch
- 分断能力 breaking capability
  - B: 16 kA H: 70 kA
  - C: 25 kA L: 85 kA
  - N: 36 kA
  - S: 50 kA
- 断路器框架电流 rated current of switch
- 脱扣器额定电流 rated current of tripping device

Remarks:

1.The design scope of this volume includes: 400V station power system-Pole 2 system;MCC of 500kV No.2 Relay Room;
2.The interface between this volume and other volumes:(1)The incoming cables of 11kV transformer are listed in D0201;(2)400V station power system -Utility System refers to D0204;(3)400V station power system -Pole 2 System refers to D0203;(4)All power cables used in this volume are listed in D0414;(5)The cable from switch cabinet to first outdoor maintenance box is designed by this volume,while cables between maintenance boxes are designed by D0405;(6)The cable from switch cabinet to outdoor lighting box is designed by this volume,while cables from outdoor lighting box to lamps and lanterns are designed by D0401 and D0402 respectively;(7)The cables from switch cabinet to secondary electrical boxes are designed by this volume,while secondary electrical boxes are designed by secondary electrical major;(8)The cables from switch cabinet to power boxes of irrigation works are designed by this volume,while power boxes are designed by manufacture;(9)The cables from switch cabinet to distribution boxes of buildings are designed by this volume,while distribution boxes are designed by other volumes respectively.
3. TN-C-S system is adopt by 400V station power system.
4. All cables designed by this volume are inflaming retarding and armoured,cable laying should according to D0406~D0410.
5.ABB switch are adopted.

材料汇总表
Material table

序号 Item No.	名称 Name	型号及规范 Specification	单位 Unit	数量 Quanity	备注 Comments
1	11kV干式配电变压器 11kV Dry type Transformer	SCZB11-2500/11，11±4x2.5%/0.4kV 2500kVA;Uk=8%;Dyn11	台 set		开列在D0204卷册 Refer to D0204
2	400V交流开关柜 400V switch cabinet	抽屉式，Drawer type	面 set	18	[深圳市泰昂能源科技股份有限公司] [Shenzhen Tieon]
3	400V交流分屏 400V MCC	固定式，Fixed type	面 set	1	[深圳市泰昂能源科技股份有限公司] [Shenzhen Tieon]
4	母线桥 Bus bridge	4000A	米 m	5	[深圳市泰昂能源科技股份有限公司] [Shenzhen Tieon]
5	母线桥 Bus bridge	1250A	米 m	5	[深圳市泰昂能源科技股份有限公司] [Shenzhen Tieon]
6	铜绞线 copper stranded conductor	150mm²	米 m	150	
7	镀锡铜鼻子 copper cable lug	DTM 150	套 set	8	
8	放热焊点 welding point	TJ150与TJ120之间 for welding of TJ150 and TJ120	套 set	8	
9	动力电缆 power cable	ZB-YJV22-0.6/1.0kV ZB-YJVP22-0.6/1.0kV	米 m		开列在D0414卷册 Refer to D0414

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

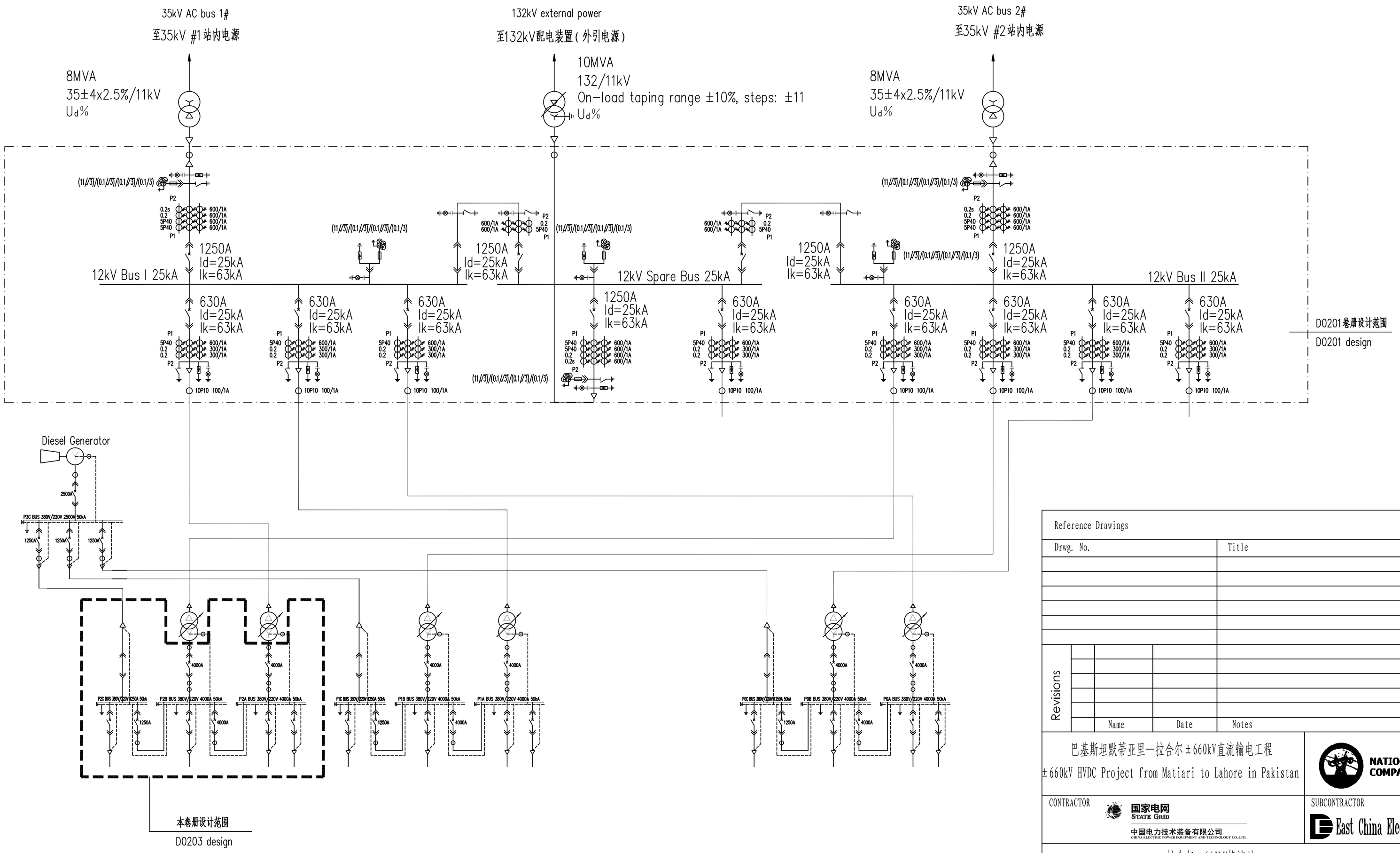
MLHVDC-LCS-ECI-DD-EC-020-001A

	姓名 Name	日期 Date
设计 Designed	雷洋	2019.05.17
校核 Checked	李桂辉	2019.05.17
审核 Reviewed	郑煜	2019.05.17
批准 Approved	潘益竿	2019.05.17

卷册设计说明及材料表
Electrical design instructions and material table

比例 Scale | 图号 DWG NO. 30-B003701S-D0203-001 | 版本 REV. A

297X420



Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

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SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: Detail Design

M L H V D C - L C S - E C I - D D - E C - 0 2 0 3 - 0 0 2 A

	姓名 Name	日期 Date
设计 Designed	雷洋	2019.05.17
校核 Checked	李建辉	2019.05.17
审核 Reviewed	郑煜	2019.05.17
批准 Approved	潘益平	2019.05.17

站用电系统接线示意图

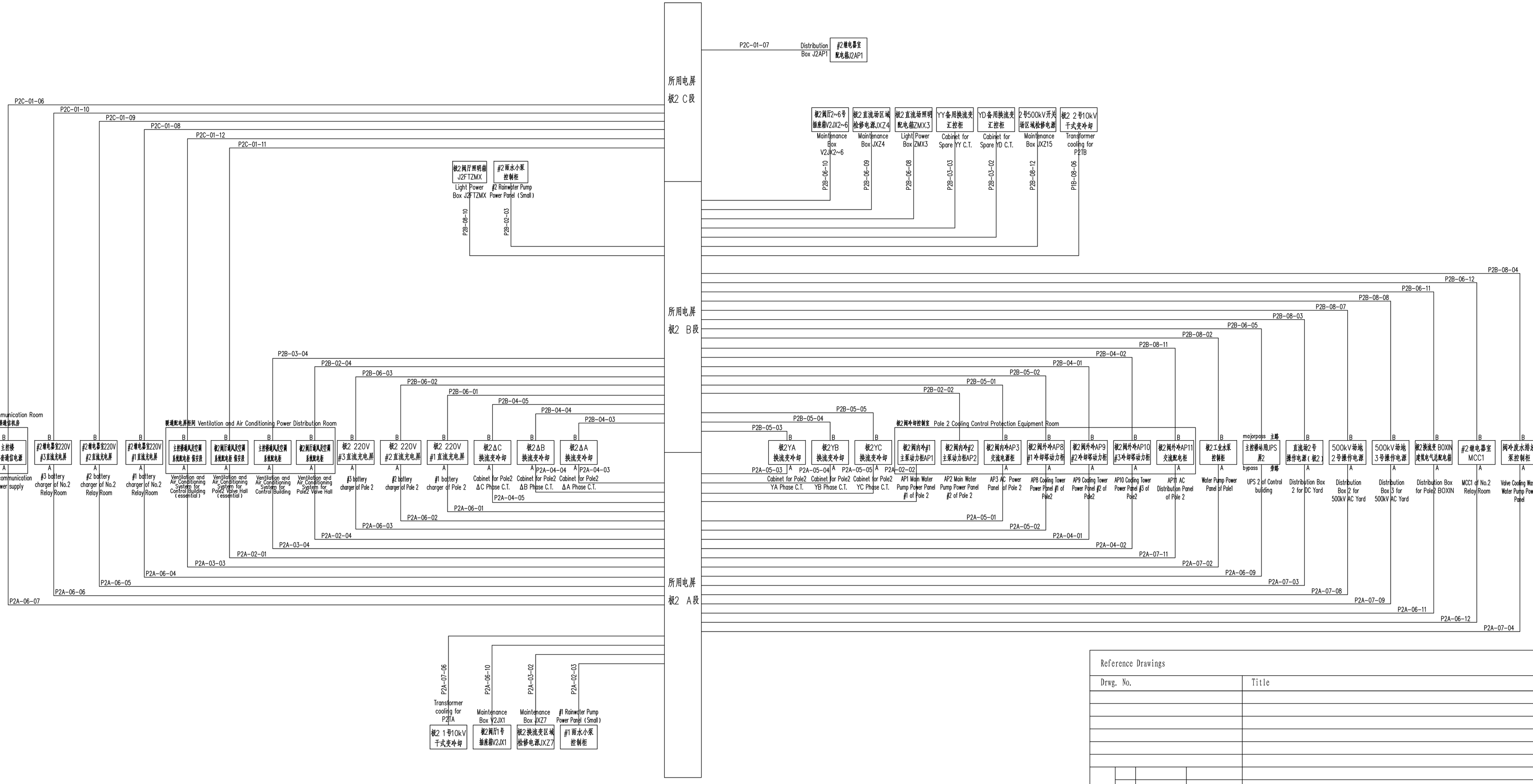
Single Line Diagram of Station Power System

比例 Scale:

图号 DWG NO.: 30-B003701S-D0203-002

版本 REV.: A

420X594



说明：1. 所有双回路供电负荷需同时在就地控制盘（柜）上设置双回路切换功能，不允许并联运行。

2. 馈线名称中A、B表示同一设备的双回电源，1、2表示不同设备。现场投切时，双电源供电的负荷应尽量平均分配在两段母线上。

Descriptions: 1. All circuits which have double-power supply should changing-over locally, and parallel operation is not permitted.

2. A and B means double-power supply of the same circuit respectively. 1 and 2 means different circuit respectively.

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - E C - 0 2 0 3 - 0 0 3 A

	姓名 Name	日期 Date
设计 Designed	雷洋	2019.05.17
校核 Checked	李建辉	2019.05.17
审核 Reviewed	郑煜	2019.05.17
批准 Approved	潘益午	2019.05.17

极2 400V站用电系统网络图

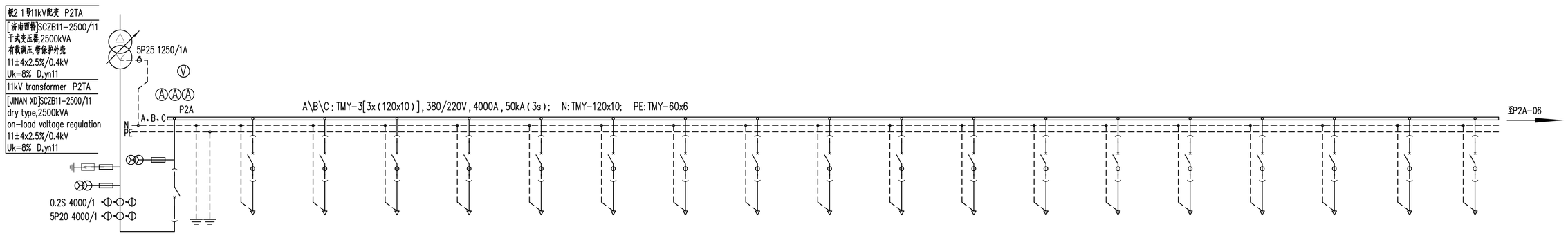
Network drawing of Pole 2 400V station power system

比例 Scale:

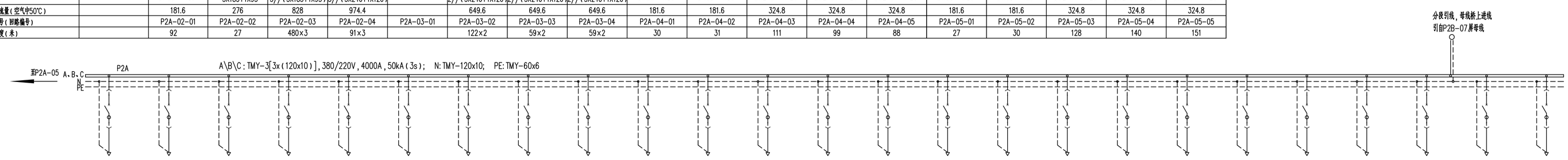
图号 DWG NO.: 30-B003701S-D0203-003

版本 REV.: A

594X841



配电屏编号		P2A-01	P2A-02					P2A-03					P2A-04					P2A-05				
屏宽(mm)		1000	650					650					650					650				
回路名称		极2 1号11kV取关进线 No.1 Pole 2 11kV transformer incoming line	极2网厅通风及空调系统配电柜 恢安柜 A Ventilation and Air Conditioning System for Pole2 Valve Hall (essential) A	极2强网内冲 #1 主泵站冲洗AP1 AP1 Main Water Pump Power Panel #1 of Pole 2	#1 雨水小泵控制柜 #1 Rainwater Pump Power Panel (Small)	极2网压通风及空调系统配电柜 A Ventilation and Air Conditioning System for Pole2 Valve Hall A	备用 for spare	极2换流变区域检修电源 JXZ7 Maintenance Box JXZ7	主控楼通风及空调系统配电柜 恢安柜 A Ventilation and Air Conditioning System for Control Building (essential) A	主控楼通风及空调系统配电柜 A Ventilation and Air Conditioning System for Control Building A	极2网冲冷AP9 #2冷却塔动力柜 A AP9 Cooling Tower Power Panel #2 of Pole2 A	极2网冲冷AP10 #3冷却塔动力柜 A AP10 Cooling Tower Power Panel #3 of Pole2 A	极2 ΔA换流变冷却 A Cabinet for Pole2 ΔA Phase C.T. A	极2 ΔB换流变冷却 A Cabinet for Pole2 ΔB Phase C.T. A	极2 ΔC换流变冷却 A Cabinet for Pole2 ΔC Phase C.T. A	极2网冲冷AP3 交流泵潜泵 A AP3 AC Power Panel of Pole2 A	极2网冲冷AP8 #1冷却塔动力柜 A AP8 Cooling Tower Power Panel #1 of Pole2 A	极2 YA换流变冷却 A Cabinet for Pole2 YA Phase C.T. A	极2 YB换流变冷却 A Cabinet for Pole2 YB Phase C.T. A	极2 YC换流变冷却 A Cabinet for Pole2 YC Phase C.T. A
回路额定容量(kW)			50	110	250	408.2			236.13	292.3	60	60	105	105	105	70	60	105	105	105	105
回路额定工作电流(A)		3798	108.2	205	443.8	734.18		~500	447.22	521	120	120	270	270	270	140	120	270	270	270	270
断路器型号		E4.2S4000 R4000 3P	XT2S160 R160	T5S400 R400	T5S630 R630	E2.2S1250 R1250	XT4S250 R250	T5S630 R630	T5S630 R630	E2.2S1250 R1250	XT2S160 R160	XT2S160 R160	T5S400 R400	T5S400 R400	T5S400 R400	XT4S250 R250	XT2S160 R160	T5S400 R400	T5S400 R400	T5S400 R400
断路器额定电流 Iu(A)		4000	160	400	630	1250	250	630	630	1250	160	160	400	400	400	250	160	400	400	400
保护电器	脱扣器型号	T-LSIG	LS/I	PR222DS/P-LSI	PR222DS/P-LSI	T-LSI	LS/I	PR222DS/P-LSI	PR222DS/P-LSI	T-LSI	LS/I	LS/I	PR222DS/P-LSI	PR222DS/P-LSI	PR222DS/P-LSI	LS/I	LS/I	PR222DS/P-LSI	PR222DS/P-LSI	PR222DS/P-LSI
	脱扣器额定电流 In(A)	4000	160	400	630	1250		630	630	1250	160	160	400	400	400	250	160	400	400	400
	瞬时无延时脱扣门限值 Ii/脱扣器电流开断值(A/s)	3900/3	120/3	240/3	630/3	875/3		630/3	630/3	625/3	144/3	144/3	320/3	320/3	320/3	160/3	144/3	320/3	320/3	320/3
	瞬时延时脱扣门限值 Ii2/脱扣器电流开断值(A/s)	12000/0.6(t=k)	400/0.1	4000/0.1	1890/0.1	5000/0.1		1890/0.1	1890/0.1	5000/0.1	1600/0.1	1600/0.1	1600/0.1	1600/0.1	1600/0.1	2000/0.1	1600/0.1	1600/0.1	1600/0.1	1600/0.1
	建议瞬动整定电流 I3(A)	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed		关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed
	单相接地延时门限值 I4/脱扣器电流开断值(A/s)	1200/0.8																			
电流互感器变比		4000/1	150/1	400/1	600/1	1200/1		600/1	600/1	1200/1	150/1	150/1	400/1	400/1	400/1	250/1	150/1	400/1	400/1	400/1
电缆型号及芯数x截面(mm²)			ZRB-YJV22-0.6/1.0 -3x95+1x50	ZRB-YJV22-0.6/1.0 -3X185+1X95	ZRB-YJV22-0.6/1.0 3//(3X185+1X95)	ZRB-YJV22-0.6/1.0 3//(3X240+1X120)		ZRB-YJV22-0.6/1.0 2//(3X240+1X120)	ZRB-YJV22-0.6/1.0 2//(3X240+1X120)	ZRB-YJV22-0.6/1.0 2//(3X240+1X120)	ZRB-YJV22-0.6/1.0 -3x95+1x50	ZRB-YJV22-0.6/1.0 -3x95+1x50	ZRB-YJV22-0.6/1.0 -3x240+1x120	ZRB-YJV22-0.6/1.0 -3x240+1x120	ZRB-YJV22-0.6/1.0 -3x240+1x120	ZRB-YJV22-0.6/1.0 -3x95+1x50	ZRB-YJV22-0.6/1.0 -3x95+1x50	ZRB-YJV22-0.6/1.0 -3x240+1x120	ZRB-YJV22-0.6/1.0 -3x240+1x120	ZRB-YJV22-0.6/1.0 -3x240+1x120
电缆载流量(空气中50℃)			181.6	276	828	974.4		649.6	649.6	649.6	181.6	181.6	324.8	324.8	324.8	181.6	181.6	324.8	324.8	324.8
电缆编号(回路编号)			P2A-02-01	P2A-02-02	P2A-02-03	P2A-02-04	P2A-03-01	P2A-03-02	P2A-03-03	P2A-03-04	P2A-04-01	P2A-04-02	P2A-04-03	P2A-04-04	P2A-04-05	P2A-05-01	P2A-05-02	P2A-05-03	P2A-05-04	P2A-05-05
电缆长度(米)			92	27	480x3	91x3		122x2	59x2	59x2	30	31	111	99	88	27	30	128	140	151



配电屏编号		P2A-06														P2A-07														
屏宽(mm)		650														1000														
回路名称		极2 220V #1直流充电屏 A #1 battery charger of Pole 2 A	极2 220V #2直流充电屏 A #2 battery charger of Pole 2 A	极2 220V #3直流充电屏 A #3 battery charger of Pole 2 A	#2储电器室220V #1直流充电屏 A #1 battery charger of No.2 Relay Room A	#2储电器室220V #2直流充电屏 A #2 battery charger of No.2 Relay Room A	#2储电器室220V #3直流充电屏 A #3 battery charger of No.2 Relay Room A	主控楼 至2#通信电源 A No.2 communication power supply A	备用 for spare	主控楼 检修UPS 2 of Control building bypass	极2网厅 检修箱 V2JX1 Maintenance Box V2JX1	极2换流变 BOXIN 换流变气体继电器 A Distribution Box for Pole2 BOXIN A	#2储电器室MCC1 A MCC1 of No.2 Relay Room A	备用 for spare	极2工业水泵控制柜 A Water Pump Power Panel of Pole2 A	直流极2号 接线电源箱(极2) A Distribution Box 2 for DC Yard A	网冲泵水箱放泵控制柜 A Valve Cooling Waste Water Pump Power Panel A	备用 for spare	极2 1号11kV 干式变冷却 Transformer cooling for P2TA	备用 for spare	500kV母线 2号接线电源 A Distribution Box 2 for 500kV AC Yard A	500kV母线 3号接线电源 A Distribution Box 3 for 500kV AC Yard A	备用 for spare	极2网冲冷AP11 交流配电柜 A AP11 AC Distribution Panel of Pole 2 A	备用 for spare	备用 for spare
回路额定容量(kW)		26.4	26.4	26.4	44	44	44	20		30(单相)		50				11		15							45			
回路额定工作电流(A)		~47	~47	~47	~79	~79	~79	~35		~145	~120	~100	~180			22	80	29.5		~12		63	63		90			
断路器型号		XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R63	XT2S160 R100	XT4S250 R250	XT2S160 R160	XT2S160 R160	XT4S250 R250	XT2S160 R100	XT2S160 R63	XT2S160 R100	XT2S160 R63	XT2S160 R63	XT2S160 R25	XT2S160 R63	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R160	XT2S160 R160	XT2S160 R160
断路器额定电流 Iu(A)		160	160	160	160	160	160	160	160	250	160	160	250	160	160	160	160	160	160	160	100	100	160	160	160	160	160
保护电器	脱扣器型号	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I
	脱扣器额定电流 In(A)	100	100	100	100	100	100	63		250	160	160	250			63	100	63		25		100	100		160			
	瞬时无延时脱扣门限值 Ii/脱扣器电流开断值(A/s)	55/3	55/3	55/3	90/3	90/3	90/3	44.1/3		160/3	144/3	120/3	200/3			63/3	90/3	44.1/3		20/3		75/3	75/3		112/3			
	瞬时延时脱扣门限值 Ii2/脱扣器电流开断值(A/s)	500/0.1	500/0.1	500/0.1	250/0.1	250/0.1	250/0.1	189/0.1		1000/0.1	800/0.1	480/0.1	500/0.1			189/0.1	250/0.1	315/0.1		关闭 Closed		200/0.1	200/0.1		1600/0.1			
	建议瞬动整定电流 I3(A)	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed		关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed			关闭 Closed	关闭 Closed	关闭 Closed		250		关闭 Closed	关闭 Closed		关闭 Closed			
	单相接地延时门限值 I4/脱扣器电流开断值(A/s)																											
电流互感器变比		100/1	100/1	100/1	100/1	100/1	100/1	75/1		250/1	150/1	150/1	250/1			75/1	100/1	75/1		30/1		100/1	100/1		150/1			
电缆型号及芯数x截面(mm²)		ZRB-YJV22-0.6/1.0 -3x16+1x10	ZRB-YJV22-0.6/1.0 -3x16+1x10	ZRB-YJV22-0.6/1.0 -3x16+1x10	ZRB-YJV22-0.6/1.0 -3x70+1x35	ZRB-YJV22-0.6/1.0 -3x70+1x35	ZRB-YJV22-0.6/1.0 -3x70+1x35	ZRB-YJV22-0.6/1.0 -3x16+1x10		ZRB-YJV22-0.6/1.0 -2x95(单A相)	ZRB-YJVP22-0.6/1.0 -3x95+1x50	ZRB-YJV22-0.6/1.0 -3x50+1x25	ZRB-YJV22-0.6/1.0 -3x150+1x70		ZRB-YJV22-0.6/1.0 -3x25+1x16	ZRB-YJV22-0.6/1.0 -3x50+1x25	ZRB-YJV22-0.6/1.0 -3x50+1x25		ZRB-YJV22-0.6/1.0 -2x10(单B相)		ZRB-YJV22-0.6/1.0 -3x50+1x25	ZRB-YJV22-0.6/1.0 -3x95+1x50		ZRB-YJV22-0.6/1.0 -3x50+1x25			
电缆载流量(空气中50℃)		63.2	63.2	63.2	149.6	149.6	149.6	63.2		188.8	181.6	121.6	240.8		83.2	121.6	121.6		53.6		121.6	181.6		121.6			
电缆编号(回路编号)		P2A-06-01	P2A-06-02	P2A-06-03	P2A-06-04	P2A-06-05	P2A-06-06	P2A-06-07	P2A-06-08	P2A-06-09	P2A-06-10	P2A-06-11	P2A-06-12	P2A-07-01	P2A-07-02	P2A-07-03	P2A-07-04	P2A-07-05	P2A-07-06	P2A-07-07	P2A-07-08	P2A-07-09	P2A-07-10	P2A-07-11	P2A-07-12	P2A-07-13
电缆长度(米)		28	32	27	368	375	367	40		60	34	122	367		219	205	233		19		302	420		30		
								注9 Note9											注9 Note9					注9 Note9			

说明:1. 两台干式配电变压器不得并列运行。
2. 站用电屏本期一次配置完成,采用抽屉式。
3. 所有馈线均应装设带电显示器,进线及分段回路装设电压表、电流表、功率表及带电显示器。
4. 有关站用电的控制、保护、信号及联闭锁要求见二次专业相关卷册。
5. 除特别注明"t=k"外,本表中断路器所选取的脱扣曲线均为反时限。
6. PR222长延时脱扣曲线时间档位所对应的电流值为@6I1,短延时脱扣曲线时间档位所对应的电流值为@8In。
7. LS/I长延时脱扣曲线时间档位所对应的电流值为@3I1,短延时脱扣曲线时间档位所对应的电流值为@10In。
8. T-LSI/T-LSIG长延时脱扣曲线时间档位所对应的电流值为@3I1,短延时脱扣曲线时间档位所对应的电流值为@10In,接地保护脱扣曲线时间档位所对应的电流值为@4In。
9. 为满足电缆热稳定要求,标注处电缆敷设长度需按图示要求,可考虑迂回敷设等方式。
10. 电流载流量根据重庆泰山电缆有限公司提供的资料确定(按空气中50°环温考虑,并考虑0.8的敷设系数)。
11. 图中开关保护整定值仅供参考,现场可根据实际情况调整。

Descriptions:1. Parallel operation of two 11kV distribution transformer is not permitted.
2. Circuit breakers in all switch cabinets are drawer type.
3. The meters should be configured accordingly.
4. The design of control and relay protection refer to secondary electrical major.
5. All tripping curve of circuit breakers are inverse time-lag type except for "t=k, which means constant time-lag.
6. The time of long delay overload tripping curve of PR222 is @6I1, the time of short delay overload tripping curve of PR222 is @8In.
7. The time of long delay overload tripping curve of LS/I is @3I1, the time of short delay overload tripping curve of LS/I is @10In.
8. The time of long delay overload tripping curve of T-LSI/T-LSIG is @3I1, the time of short delay overload tripping curve of T-LSI/T-LSIG is @10In, the time of ground protection tripping curve of T-LSIG is @4In.
9. The length of cables should be laid according to the drawing.
10. Current-carrying capacity of cable is according to cable manufacture.
11. The settings of tripping device are for reference.

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

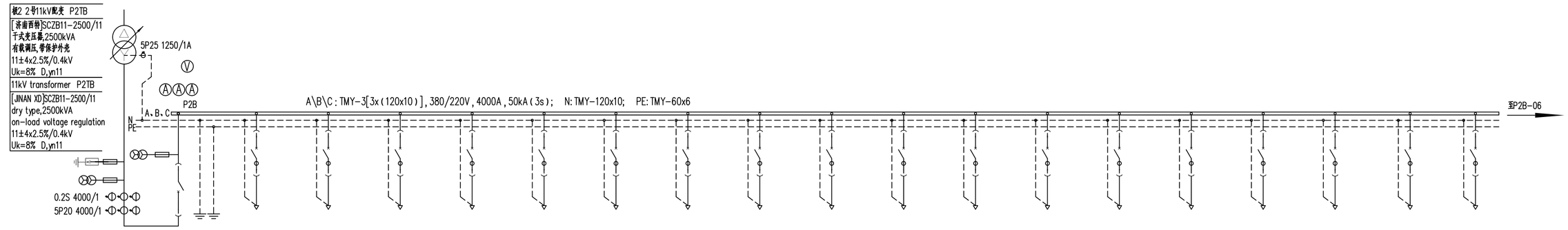
M L H V D C - L C S - E C I - D D - E C - 0 2 0 3 - 0 0 4 A

	姓名 Name	日期 Date
设计 Designed	雷洋	2019.05.17
校核 Checked	李志辉	2019.05.17
审核 Reviewed	郑煜	2019.05.17
批准 Approved	潘益竹	2019.05.17

极2 400V站用电系统P2A段配置图
Configuration drawing of P2A section of 400V Station Power System

比例 Scale | 图号 DWG NO. 30-B003701S-D0203-004 | 版本 REV. A

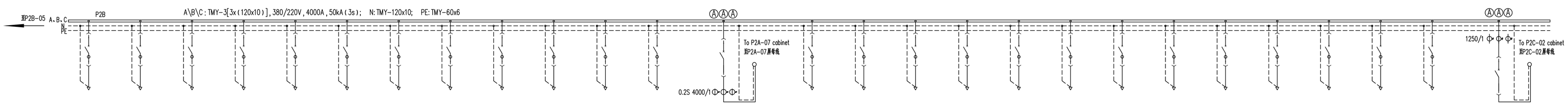
594X841



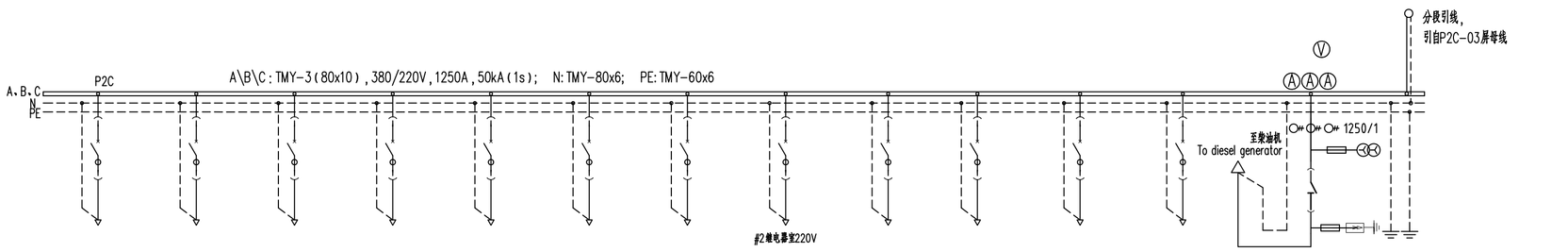
配电屏编号	P2B-01	P2B-02				P2B-03				P2B-04				P2B-05			
屏宽(mm)	1000	650				650				650				650			
回路名称	极2 2号11kV配变进线 No.2 Pole 2 11kV transformer incoming line	备用 for spare	极2泵房内冷 #2主泵缺油报AP2 AP2 Main Water Pump Power Panel #2 of Pole 2	极2雨水小泵控制柜 #2 Rainwater Pump Power Panel (Small)	极2网厅通风及空调系统配电柜B Ventilation and Air Conditioning System for Pole2 Valve Hall B	备用 for spare	YD备用换流变 Z变柜 Cabinet for Spare YD C.T.	YY备用换流变 Z变柜 Cabinet for Spare YY C.T.	主控楼通风及空调系统配电柜B Ventilation and Air Conditioning System for Control Building B	极2冷却塔AP9 #2冷却塔动力柜B AP9 Cooling Tower Power Panel #2 of Pole2 B	极2冷却塔AP10 #3冷却塔动力柜B AP10 Cooling Tower Power Panel #3 of Pole2 B	极2 ΔA换流变冷却B Cabinet for Pole2 ΔA Phase C.T. B	极2 ΔB换流变冷却B Cabinet for Pole2 ΔB Phase C.T. B	极2 ΔC换流变冷却B Cabinet for Pole2 ΔC Phase C.T. B	极2网内冷AP3 交流电源柜B AP3 AC Power Panel of Pole2 B	极2网内冷AP8 #1冷却塔动力柜B AP8 Cooling Tower Power Panel #1 of Pole2 B	极2 YA换流变冷却B Cabinet for Pole2 YA Phase C.T. B	极2 YB换流变冷却B Cabinet for Pole2 YB Phase C.T. B	极2 YC换流变冷却B Cabinet for Pole2 YC Phase C.T. B
回路额定容量(kW)			110	250	408.2		105	105	292.3	60	60	105	105	105	70	60	105	105	105
回路额定工作电流(A)	3798		205	443.8	734.18		270	270	521	120	120	270	270	270	140	120	270	270	270
断路器型号	E4.2S4000 R4000 3P	XT4S250 R250	T5S400 R400	T5S630 R630	E2.2S1250 R1250	XT4S250 R250	T5S400 R400	T5S400 R400	E2.2S1250 R1250	XT2S160 R160	XT2S160 R160	T5S400 R400	T5S400 R400	T5S400 R400	XT4S250 R250	XT2S160 R160	T5S400 R400	T5S400 R400	T5S400 R400
断路器额定电流 Iu(A)	4000	250	400	630	1250	250	400	400	1250	160	160	400	400	400	250	160	400	400	400
保护电器 脱扣器型式	T-LSIG	LS/I	PR222DS/P-LSI	PR222DS/P-LSI	T-LSI	LS/I	PR222DS/P-LSI	PR222DS/P-LSI	T-LSI	LS/I	LS/I	PR222DS/P-LSI	PR222DS/P-LSI	PR222DS/P-LSI	LS/I	LS/I	PR222DS/P-LSI	PR222DS/P-LSI	PR222DS/P-LSI
保护电器 脱扣器额定电流 In(A)	4000		400	630	1250		400	400	1250	160	160	400	400	400	250	160	400	400	400
保护电器 瞬时或短延时脱扣整定值I1/脱扣器电流比倍数(A/s)	3900/3		240/3	630/3	875/3		320/3	320/3	625/3	144/3	144/3	320/3	320/3	320/3	160/3	144/3	320/3	320/3	320/3
保护电器 瞬时或短延时脱扣整定值I2/脱扣器电流比倍数(A/s)	12000/0.6(t=k)		4000/0.1	1890/0.1	5000/0.1		1200/0.1	1200/0.1	5000/0.1	1600/0.1	1600/0.1	1600/0.1	1600/0.1	1600/0.1	2000/0.1	1600/0.1	1600/0.1	1600/0.1	1600/0.1
保护电器 接地保护整定电流I3(A)	关闭 Closed		关闭 Closed	关闭 Closed	关闭 Closed				关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed
保护电器 接地保护整定值I4/脱扣器电流比倍数(A/s)	1200/0.8																		
电流互感器变比	4000/1		400/1	600/1	1200/1		400/1	400/1	1200/1	150/1	150/1	400/1	400/1	400/1	250/1	150/1	400/1	400/1	400/1
电缆型号及芯数x截面(mm²)			ZRB-YJV22-0.6/1.0 -3X185+1X95	ZRB-YJV22-0.6/1.0 3//(3X185+1X95)	ZRB-YJV22-0.6/1.0 3//(3X240+1X120)		ZRB-YJV22-0.6/1.0 -3x240+1x120	ZRB-YJV22-0.6/1.0 -3x240+1x120	ZRB-YJV22-0.6/1.0 2//(3X240+1X120)	ZRB-YJV22-0.6/1.0 -3x95+1x50	ZRB-YJV22-0.6/1.0 -3x95+1x50	ZRB-YJV22-0.6/1.0 -3x240+1x120	ZRB-YJV22-0.6/1.0 -3x240+1x120	ZRB-YJV22-0.6/1.0 -3x240+1x120	ZRB-YJV22-0.6/1.0 -3x95+1x50	ZRB-YJV22-0.6/1.0 -3x95+1x50	ZRB-YJV22-0.6/1.0 -3x240+1x120	ZRB-YJV22-0.6/1.0 -3x240+1x120	ZRB-YJV22-0.6/1.0 -3x240+1x120
电缆载流量(空气中50℃)			400	828	974.4		324.8	324.8	649.6	181.6	181.6	324.8	324.8	324.8	181.6	181.6	324.8	324.8	324.8
电缆编号(回路编号)		P2B-02-01	P2B-02-02	P2B-02-03	P2B-02-04	P2B-03-01	P2B-03-02	P2B-03-03	P2B-03-04	P2B-04-01	P2B-04-02	P2B-04-03	P2B-04-04	P2B-04-05	P2B-05-01	P2B-05-02	P2B-05-03	P2B-05-04	P2B-05-05
电缆长度(米)			36	465x3	98x3		158	257	63x2	39	40	103	92	80	35	42	120	132	143

说明:1. 两台干式配电变压器不得并列运行。
2. 站用电屏本期一次配置完成,采用抽屉式。
3. 所有馈线均应装设带电显示器,进线及分段回路装设电压表、电流表、功率表及带电显示器。
4. 有关站用电的控制、保护、信号及联闭锁要求见二次专业相关卷册。
5. 除特别注明"t=k"外,本表中断路器所选取的脱扣曲线均为反时限。
6. PR222长延时脱扣曲线时间档位所对应的电流值为@6I1,短延时脱扣曲线时间档位所对应的电流值为@8In。
7. LS/I长延时脱扣曲线时间档位所对应的电流值为@3I1,短延时脱扣曲线时间档位所对应的电流值为@10In。
8. T-LSI/T-LSIG长延时脱扣曲线时间档位所对应的电流值为@3I1,短延时脱扣曲线时间档位所对应的电流值为@10In,接地保护脱扣曲线时间档位所对应的电流值为@4In。
9. 为满足电缆热稳定要求,标注处电缆敷设长度需按图示要求,可考虑迂回敷设等方式。
10. 电流载流量根据重庆泰山电缆有限公司提供的资料确定(按空气中50°环温考虑,并考虑0.8的敷设系数)。
11. 图中开关保护整定值仅供参考,现场可根据实际情况调整。

Descriptions:1. Parallel operation of two 11kV distribution transformer is not permitted.
2. Circuit breakers in all switch cabinets are drawer type.
3. The meters should be configured accordingly.
4. The design of control and relay protection refer to secondary electrical major.
5. All tripping curve of circuit breakers are inverse time-lag type except for "t=k", which means constant time-lag.
6. The time of long delay overload tripping curve of PR222 is @6I1, the time of short delay overload tripping curve of PR222 is @8In.
7. The time of long delay overload tripping curve of LS/I is @3I1, the time of short delay overload tripping curve of LS/I is @10In.
8. The time of long delay overload tripping curve of T-LSI/T-LSIG is @3I1, the time of short delay overload tripping curve of T-LSI/T-LSIG is @10In, the time of ground protection tripping curve of T-LSIG is @4In.
9. The length of cables should be laid according to the drawing.
10. Current-carrying capacity of cable is according to cable manufacture.
11. The settings of tripping device are for reference.



配电屏编号	P2B-06													P2B-07	P2B-08												P2C-03
屏宽(mm)	650													1000	650												1000
回路名称	极2 220V #1直流充电屏B #1 battery charger of Pole 2 B	极2 220V #2直流充电屏B #2 battery charger of Pole 2 B	极2 220V #3直流充电屏B #3 battery charger of Pole 2 B	备用 for spare	主控楼 站用UPS极2 主楼 UPS 2 of Control building mojoroass	备用 for spare	备用 for spare	极2 直流分屏柜 配电盘 ZM03 Light Power Box ZM03	极2直流场 区域测量电表 JXZ4 Maintenance Box JXZ4	极2网厅2-6号 触摸屏 V2.0U2-6 Maintenance Box V2.0U2-6	极2换流变BOXN 建筑电气区配电盘B Distribution Box for Pole2 BOXN B	#2继电器室MCC1 B MCC1 of No.2 Relay Room B	分段 section switch	备用 for spare	极2工业水泵控制柜B Water Pump Power Panel of Pole2 B	直流场2号 器控电源(极2) B Distribution Box 2 for DC Yard B	网外脱水排放泵控制柜B Valve Cooling Waste Water Pump Power Panel B	备用 for spare	极2 2号11kV 干式变压器 Transformer cooling for P2TB	500kV场地 2号器控电源 B Distribution Box 2 for 500kV AC Yard B	500kV场地 3号器控电源 B Distribution Box 3 for 500kV AC Yard B	备用 for spare	极2网厅照明箱 J2FTZMX Light Power Box J2FTZMX	极2网外冷AP11 交流配电柜B AP11 AC Distribution Panel of Pole 2 B	2号500kV开关场 区域测量电表 JXZ15 Maintenance Box JXZ15	备用 for spare	柴油机与工作母线分段 section switch
回路额定容量(kW)	26.4	26.4	26.4		30						50					11	15						20.42	45			
回路额定工作电流(A)	~47	~47	~47		~54			40	~160	~160	~100	~160		3798		22	80	29.5		~12	63	63		53.06	90	~160		
断路器型号	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT4S250 R250	XT4S250 R250	XT2S160 R160	XT4S250 R250		E4.2S4000 R4000 3P	XT2S160 R100	XT2S160 R63	XT2S160 R100	XT2S160 R63	XT2S160 R100	XT2S160 R25	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R160	XT4S250 R250	E2.2S1250 R1250 3P
断路器额定电流 Iu(A)	160	160	160	160	160	160	160	160	250	250	160	250		4000	160	160	160	160	160	160	160	160	160	160	160	250	1250
保护电器 脱扣器型式	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I		T-LSI	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	T-LSI
保护电器 脱扣器额定电流 In(A)	100	100	100		100			100	250	250	160	250		4000		63	100	63		25	100	100		100	160	250	1250
保护电器 瞬时或短延时脱扣整定值I1/脱扣器电流比倍数(A/s)	55/3	55/3	55/3		70/3			50/3	160/3	160/3	120/3	200/3		3900/3		63/3	90/3	44/3		20/3	75/3	75/3		60/3	112/3	160/3	1250/3
保护电器 瞬时或短延时脱扣整定值I2/脱扣器电流比倍数(A/s)	500/0.1	500/0.1	500/0.1		300/0.1			200/0.1	500/0.1	500/0.1	480/0.1	500/0.1		10000/0.4(t=k)		189/0.1	250/0.1	315/0.1		关闭 Closed	200/0.1	200/0.1		300/0.1	1600/0.1	500/0.1	10000/0.4(t=k)
保护电器 建议瞬动整定电流I3(A)	关闭 Closed	关闭 Closed	关闭 Closed		关闭 Closed			关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed		关闭 Closed		关闭 Closed	关闭 Closed	关闭 Closed		250	关闭 Closed	关闭 Closed		关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed
保护电器 建议接地保护整定值I4/脱扣器电流比倍数(A/s)														1200/0.8													500/0.4
电流互感器变比	100/1	100/1	100/1		100/1			100/1	250/1	250/1	150/1	250/1		4000/1		75/1	100/1	75/1		30/1	100/1	100/1		100/1	150/1	250/1	1250/1
电缆型号及芯数x截面(mm²)	ZRB-YJV22-0.6/1.0 -3x16+1x10	ZRB-YJV22-0.6/1.0 -3x16+1x10	ZRB-YJV22-0.6/1.0 -3x16+1x10		ZRB-YJV22-0.6/1.0 -3x35+1x16			ZRB-YJV22-0.6/1.0 -3x16+1x10	ZRB-YJV22-0.6/1.0 -3x95+1x50	ZRB-YJV22-0.6/1.0 -3x95+1x50	ZRB-YJV22-0.6/1.0 -3x50+1x25	ZRB-YJV22-0.6/1.0 -3x150+1x70	母线桥			ZRB-YJV22-0.6/1.0 -3x25+1x16	ZRB-YJV22-0.6/1.0 -3x50+1x25	ZRB-YJV22-0.6/1.0 -3x50+1x25		ZRB-YJV22-0.6/1.0 -2x10(接头相)	ZRB-YJV22-0.6/1.0 -3x50+1x25	ZRB-YJV22-0.6/1.0 -3x95+1x50		ZRB-YJV22-0.6/1.0 -3x25+1x16	ZRB-YJV22-0.6/1.0 -3x50+1x25	ZRB-YJV22-0.6/1.0 -3x185+1x95		母线桥
电缆载流量(空气中50℃)	63.2	63.2	63.2		100			63.2	181.6	181.6	121.6	240.8				83.2	121.6	121.6		53.6	121.6	181.6		83.2	121.6	276	
电缆编号(回路编号)	P2B-06-01	P2B-06-02	P2B-06-03	P2B-06-04	P2B-06-05	P2B-06-06	P2B-06-07	P2B-06-08	P2B-06-09	P2B-06-10	P2B-06-11	P2B-06-12			P2B-08-01	P2B-08-02	P2B-08-03	P2B-08-04	P2B-08-05	P2B-08-06	P2B-08-07	P2B-08-08	P2B-08-09	P2B-08-10	P2B-08-11	P2B-08-12	P2B-08-13	
电缆长度(米)	37	34	38		68			109	110	56	115	374				212	208	240		19	309	428		64	40	252		
	注9 Note9	注9 Note9	注9 Note9																	注9 Note9							



配电屏编号	P2C-01											P2C-02
屏宽(mm)	650											1000
回路名称	备用 for spare	备用 for spare	备用 for spare	备用 for spare	备用 for spare	主控楼 #2通信电源B No.2 communication power supply B	#2继电器室 配电盘J2AP1 Distribution Box J2AP1	#2继电器室220V #1直流充电屏B #1 battery charger of No.2 Relay Room B	#2继电器室220V #2直流充电屏B #2 battery charger of No.2 Relay Room B	#2继电器室220V #3直流充电屏B #3 battery charger of No.2 Relay Room B	极2网厅通风及空调系统配电柜 保安段B Ventilation and Air Conditioning System for Pole2 Valve Hall (essential) B	主控楼通风及空调系统配电柜 保安段B Ventilation and Air Conditioning System for Control Building (essential) B	柴油机进线 incoming line of diesel generator
回路额定容量(kW)						20	28	44	44	44	50	236.13	466.13
回路额定工作电流(A)						~35	43	~79	~79	~79	108.2	447.22	870.42
断路器型号	XT2S160 R63	XT2S160 R63	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R63	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R160	T5S630 R630	隔离开关 disconnector
断路器额定电流 Iu(A)	160	160	160	160	160	160	160	160	160	160	160	630	E2.2S/MS/1250/R1250/3P
保护电器 脱扣器型式	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	PR222DS/P-LSI	
保护电器 脱扣器额定电流 In(A)						63	100	100	100	100	160	630	
保护电器 瞬时或短延时脱扣整定值I1/脱扣器电流比倍数(A/s)						44.1/3	90/3	90/3	90/3	90/3	120/3	630/3	
保护电器 瞬时或短延时脱扣整定值I2/脱扣器电流比倍数(A/s)						189/0.1	200/0.1	250/0.1	250/0.1	250/0.1	400/0.1	1890/0.1	
保护电器 建议瞬动整定电流I3(A)						关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed	
保护电器 建议接地保护整定值I4/脱扣器电流比倍数(A/s)													
电流互感器变比						75/1	100/1	100/1	100/1	100/1	150/1	600/1	
电缆型号及芯数x截面(mm²)						ZRB-YJV22-0.6/1.0 -3x16+1x10	ZRB-YJV22-0.6/1.0 -3x50+1x25	ZRB-YJV22-0.6/1.0 -3x70+1x35	ZRB-YJV22-0.6/1.0 -3x70+1x35	ZRB-YJV22-0.6/1.0 -3x70+1x35	ZRB-YJV22-0.6/1.0 -3x95+1x50	ZRB-YJV22-0.6/1.0 2//(3X240+1X120)	ZRB-YJV22-0.6/1.0 4//(3X240+1X120)
电缆载流量(空气中50℃)						63.2	121.6	149.6	149.6	149.6	181.6	649.6	1299.2
电缆编号(回路编号)	P2C-01-01	P2C-01-02	P2C-01-03	P2C-01-04	P2C-01-05	P2C-01-06	P2C-01-07	P2C-01-08	P2C-01-09	P2C-01-10	P2C-01-11	P2C-01-12	P2C-02-01
电缆长度(米)						38	394	370	377	369	88	55x2	208x4
						注9 Note9							

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - E C - 0 2 0 3 - 0 0 5 A

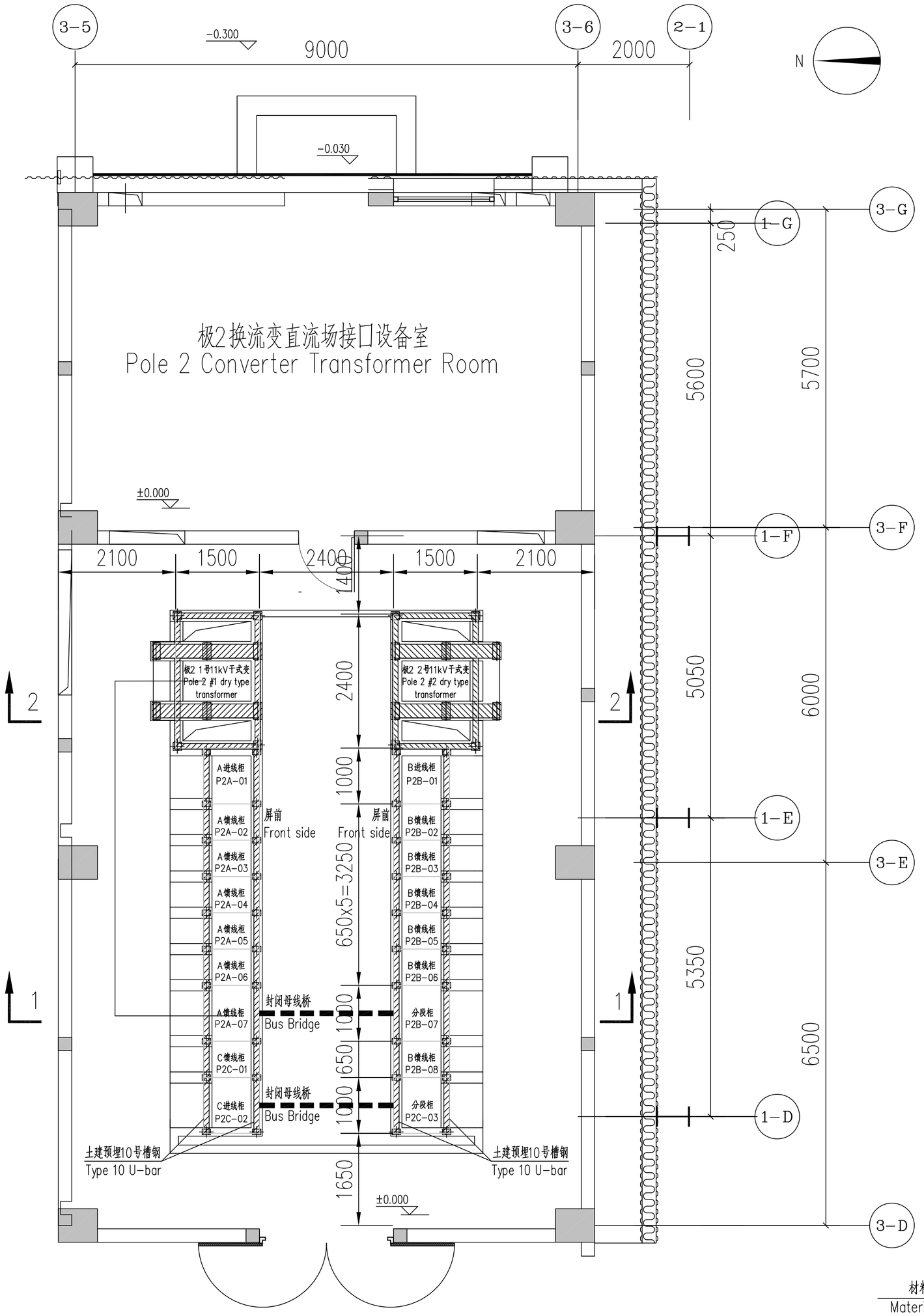
	姓名 Name	日期 Date
设计 Designed	雷洋	2019.05.17
校核 Checked	凌艳辉	2019.05.17
审核 Reviewed	郑煜	2019.05.17
批准 Approved	潘益午	2019.05.17

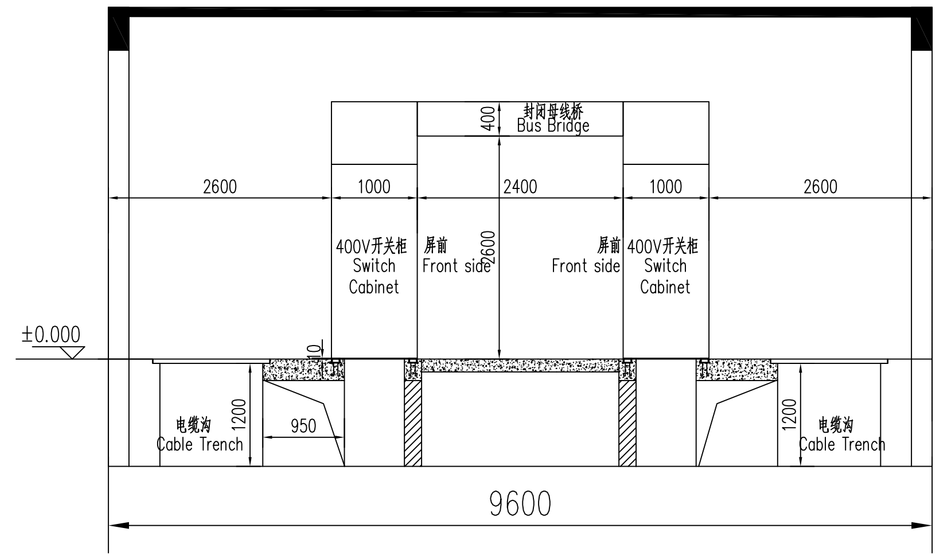
极2 400V站用电系统P2B及P2C段配置图
Configuration drawing of P2B and P2C sections of 400V Station Power System

比例 Scale | 图号 DWG NO. 30-B003701S-D0203-005 | 版本 REV. A

594X841

极2 380/220V交流阀组配电室
Pole 2 380/220V AC transformer room





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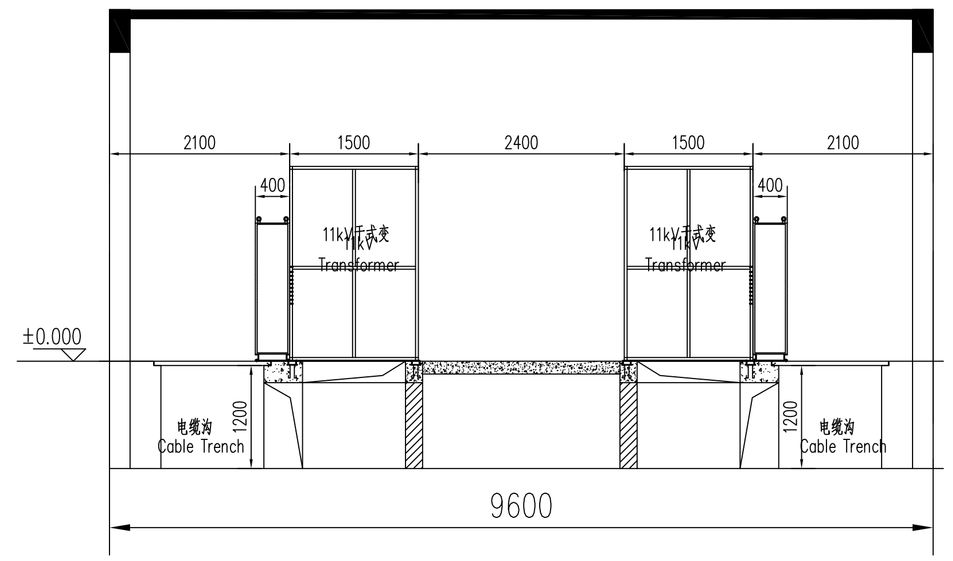

2-2

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
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拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - E C - 0 2 0 3 - 0 0 6 A

	姓名 Name	日期 Date
设计 Designed	雷洋	2019.05.17
校核 Checked	李建辉	2019.05.17
审核 Reviewed	郑煜	2019.05.17
批准 Approved	潘益竿	2019.05.17

极2 400V站用电室布置图
Layout Drawing of Pole 2 400V Station Service House

比例 Scale: | 图号 DWG NO.: 30-B003701S-D0203-006 | 版本 REV.: A

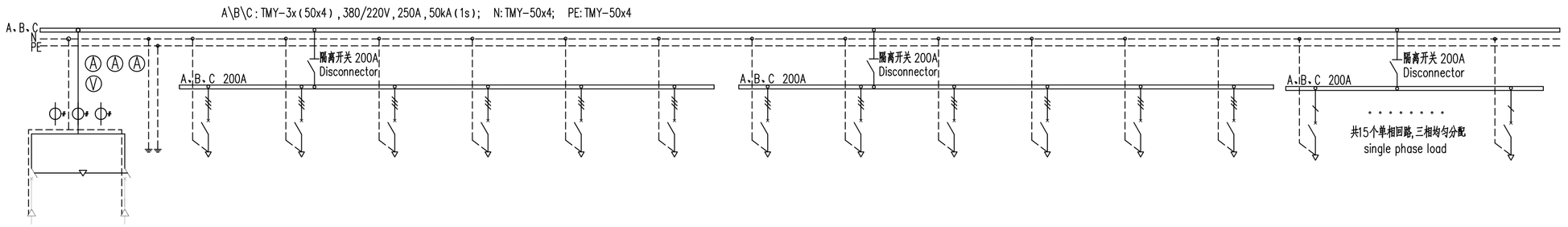
说明：1. 400V开关柜与屏柜底部预埋槽钢螺栓连接。
2. 封闭母线桥柜顶走线，馈线电缆下出线。
3. 400V开关柜PE线和N线在P2A-01、P2B-01、P2C-02屏处采用TJ-150铜绞线与主地网连接，三处接至主地网不同干线，接地材料开列在本图。
4. 11kV干式变采用TJ-150铜绞线与基础下方预留接地引上（规格：50x6镀锌扁钢）连接。

Remarks：1.The cabinet should be connected to U-bar with bolts.
2. Bus bridge are arranged on top of cabinet, cables are extracted from the bottom of cabinet.
3. The PE and N line of P2A-01、P2B-01、P2C-02 cabinet should be connected to the grounding grid..
4. 50x6 steel are reserved at foundation of transformer for transformer grounding.

材料表
Material table

序号 Item No.	名称 Name	型号及规范 Specification	单位 Unit	数量 Quanity	备注 Comments
1	铜绞线 copper stranded conductor	150mm²	米 m	120	
2	镀锡铜鼻子 copper cable lug	DTM 150	套 set	6	
3	放热焊点 welding point	TJ150与TJ120之间 for welding of TJ150 and TJ120	套 set	6	

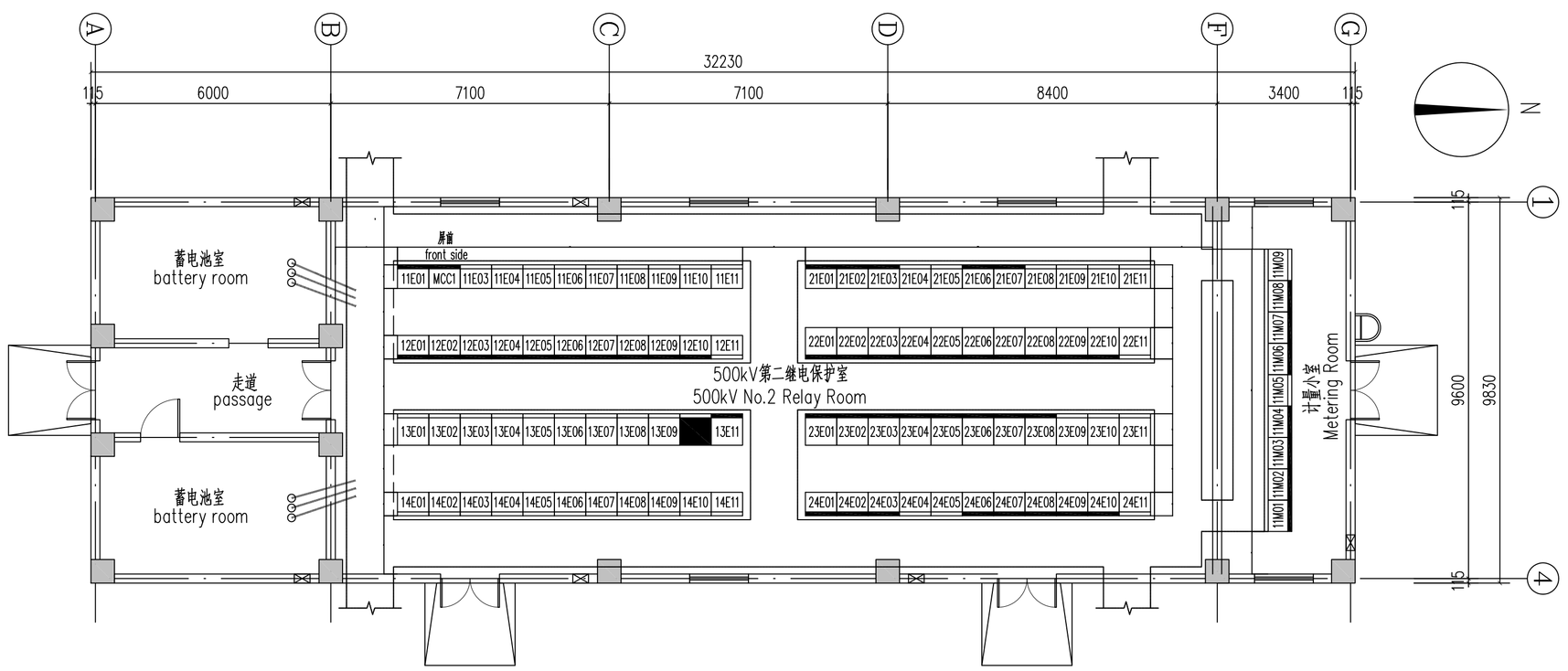
594X841



配电屏编号			#2继电器室MCC1														
屏宽(mm)			800														
回路名称	进线		#2继电器室配电箱J2AL1 Distribution Box J2AL1	备用 for spare	备用 for spare	备用 for spare	备用 for spare	500kV开关场户外照明配电箱5 Light Power Box HWZM5	#2继电器室试验电源屏1 No.1 testing power pannel	#2继电器室试验电源屏2 No.2 testing power pannel	#2继电器室事故照明屏 Emergency Lighting pannel	备用 for spare	备用 for spare	备用 for spare	二次单相回路1 single phase load 1	二次单相回路n single phase load n	二次单相回路15 single phase load 15
回路额定容量(kW)			22					17.56	15	15	5kVA(单相)						
回路额定工作电流(A)	~180		34					40.1	30	30	25						
断路器型号	ATS-CB010/T3S250 R250		XT2S160 R63	XT2S160 R63	XT2S160 R63	XT2S160 R63	XT2S160 R63	XT2S160 R100	XT2S160 R63	XT2S160 R63	XT2S160 R100	XT2S160 R100	XT2S160 R100	XT2S160 R100	S201-C32	S201-C32	S201-C32
断路器额定电流Iu(A)	250	250	160	160	160	160	160	160	160	160	160	160	160	160			
保护脱扣器 脱扣器型号	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I	LS/I			
保护脱扣器 脱扣器额定电流In(A)	250	250	63					100	63	63	63						
保护脱扣器 建议长延时脱扣门限值I1/脱扣曲线时间档位(A/s)	200/3	200/3	50.4/3					60/3	37.8/3	37.8/3	37.8/3						
保护脱扣器 建议短延时脱扣门限值I2/脱扣曲线时间档位(A/s)	500/0.1	500/0.1	126/0.1					200/0.1	126/0.1	126/0.1	126/0.1						
保护脱扣器 建议瞬动整定电流I3(A)	关闭 Closed	关闭 Closed	关闭 Closed					关闭 Closed	关闭 Closed	关闭 Closed	关闭 Closed						
保护脱扣器 建议接地保护门限值I4/脱扣曲线时间档位(A/s)																	
电流互感器变比	250/1	250/1	75/1					100/1	75/1	75/1	75/1						
相数															共15个单相回路,三相均匀分配 15 single phase load		
电缆型号及芯数x截面(mm²)			ZRB-YJV22-0.6/1.0 -3x16+1x10					ZRB-YJV22-0.6/1.0 -3x25+1x16	ZRB-YJV22-0.6/1.0 -4x10	ZRB-YJV22-0.6/1.0 -4x10	ZRB-YJV22-0.6/1.0 -2x10(接A相)						
电缆载流量(空气中50℃)			63.2					83.2	47.2	47.2	53.6						
电缆编号(回路编号)	来自站用电屏P2A-06-12	来自站用电屏P2B-06-12	J2-01-01	J2-01-02	J2-01-03	J2-01-04	J2-01-05	J2-01-06	J2-01-07	J2-01-08	J2-01-09	J2-01-10	J2-01-11	J2-01-12			
电缆长度(米)			35					63	21	21	9						

说明:1. #2继电器室MCC1交流分屏位置如图所示。
2. 交流分屏本期一次配置完成,采用固定式。
3. 交流分屏馈线中的二次单相回路电缆由二次专业设计并开列,见二次专业相关图纸。
4. LS/I长延时脱扣曲线时间档位所对应的电流值为@3I1,短延时脱扣曲线时间档位所对应的电流值为@10In。
5. 交流分屏MCC1与基础预埋槽钢焊接安装,N线、PE线分别通过TJ-150铜绞线与户外主地网放热焊接。
6. 电流载流量根据重庆泰山电缆有限公司提供的资料确定(按空气中50°环温考虑,并考虑0.8的敷设系数)。
7. 图中开关保护整定值仅供参考,现场可根据实际情况调整。

Descriptions: 1. The location of MCC1 is shown as Layout drawing of 500kV No.2 Relay Room.
2. Circuit breakers in MCC1 are fixed type.
3. The single phase circuits are designed by the secondary electrical major .
4. The time of short delay overload tripping curve of LS/I is @3I1.
The time of short delay overload tripping curve of LS/I is @10In.
5. Switch cabinet MCC1 should be welded to U-bar of the foundation when installing.
6. Current-carrying capacity of cable is according to cable manufacture.
7. The settings of tripping device are for reference.



#2继电器室MCC1交流分屏布置图

Layout drawing of 500kV No.2 relay room

材料表

Material table

序号 Item No.	名称 Name	型号及规范 Specification	单位 Unit	数量 Quanity	备注 Comments
1	铜绞线 copper stranded conductor	150mm²	米 m	30	
2	镀锡铜鼻子 copper cable lug	DTM 150	套 set	2	
3	放热焊点 welding point	TJ150与TJ120之间 for welding of TJ150 and TJ120	套 set	2	

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

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SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - E C - 0 2 0 3 - 0 0 7 A

	姓名 Name	日期 Date
设计 Designed	雷洋	2019.05.17
校核 Checked	李艳辉	2019.05.17
审核 Reviewed	郑煜	2019.05.17
批准 Approved	潘益竹	2019.05.17

#2继电器室MCC1交流分屏配置及布置图

Configuration and Layout Drawing MCC1 of 500kV No.2 Relay Room

比例 Scale | 图号 DWG NO. 30-B003701S-D0203-007 | 版本 REV. A

594X841